0.1 Preliminaries

Recall that:

Definition 0.1.1. Let $(X, \mathcal{T})$ be a Topological Space.
A map $f : \mathbb{N} \to X : n \mapsto x_n$ is called a *sequence* in X.

Definition 0.1.2. Let $(X, \mathcal{T})$ be a Topologcial Space, and $\{x_n\}$ be a sequence in X.
$\{x_n\}$ is said to be *converges* to $x \in X$ if:

For any open neighborhood $U \in \mathcal{T}$ of x, there exists $N \in \mathbb{N}$ such that $\forall n \geq N$, $x_n \in U$

Note: Generally, the limit of a sequence is Not unique.

Theorem 0.1.3. Let $f : X \to Y$ be a continuous map.
If a sequence $\{x_n\}$ in X converges to $x \in X$, then $\{f(x_n)\}$ converges to $f(x)$.

Theorem 0.1.4. Let X be a Hausdorff Space. Then, Every convergent sequence has a unique limit.

0.2 First-Countable

Definition 0.2.1. Let $(X, \mathcal{T})$ be a Topological Space, and let $x \in X$ be given.
A collection $\beta_x \subset \mathcal{T}$ is called a *Local Basis* at x if it satisfies:

1. For any $B \in \beta_x$, $x \in B$.
2. For any open neighborhood U_x of x, there exists $B \in \beta_x$ such that $x \in B \subseteq U_x$

Note: every Topolgocai Space has a local basis at any point. More specifically,

$$\beta_x = \{B \in \mathcal{T} \mid x \in B\}$$

is local basis at x. But, there is a 'good' local basis only under some condition.

Definition 0.2.2. A Topological Space X is called *First-Countable* if:

For any $x \in X$, there exists a Countable local basis at x

Theorem 0.2.3. Suppose that $(X, \mathcal{T}_X)$ is a First-Countable Space, and $(Y, \mathcal{T}_Y)$ is any Topolgocial Space. TFAE:

1. A map $f : X \to Y$ is Continuous.
2. If a sequence $\{x_n\}$ in X converges to $x \in X$, then $\{f(x_n)\}$ converges to $f(x)$.

Theorem 0.2.4. Suppose that $(X, \mathcal{T})$ is a First-Countable Space. TFAE:

1. X is Hausdorff Space.
2. Every convergent sequcne in X has a unique limit.